P-Values Explained

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Introduction

The p-value is the single most reported statistic in applied research, and the most frequently misinterpreted. Understanding what a p-value is – and what it is not – is essential for reading, writing, and reviewing quantitative studies. The American Statistical Association's 2016 statement on p-values laid out six principles that every user of statistics should internalise.

Prerequisites

Basic hypothesis testing and sampling distributions.

Theory

A **p-value** is the probability, assuming the null hypothesis is true, of observing data at least as extreme as what was actually observed. Formally,

$$p = P(T \geq t_{\text{obs}} \mid H_0)$$

for a one-sided test with test statistic T, or a two-sided analogue for two-sided alternatives.

Correct interpretation: a small p-value means the observed data would be unusual under H_0. It does **not** mean that H_0 is unlikely to be true, nor that the alternative is probable.

Common misinterpretations:

1. "$p = 0.03$ means there's a 3% probability the null is true." Wrong: the p-value is a conditional probability of data given H_0, not of H_0 given data.
2. "$p < 0.05$ means the result is meaningful." Wrong: small p-values can come from huge samples detecting trivial effects.
3. "$p > 0.05$ means no effect." Wrong: failure to reject is not evidence of no effect; it is absence of evidence.
4. "p-values from the same experiment are comparable." Wrong: p-values are discontinuous in the effect size; a p of 0.04 and 0.06 are nearly identical evidentially.

Under H_0, the p-value has a uniform distribution on $[0, 1]$ (for continuous test statistics). This is a useful simulation check.

Assumptions

- The null hypothesis and test statistic are pre-specified.
- The sampling model under H_0 is correctly specified.
- Multiple testing, optional stopping, and selective reporting are not present.

R Implementation

```r
library(ggplot2)
set.seed(2026)
```

```r
# Simulate p-values under H0 (they should be uniform)
n_sims <- 5000
pvals_null <- replicate(n_sims, {
  x <- rnorm(30, mean = 0, sd = 1)   # H0 true: mu = 0
  t.test(x, mu = 0)$p.value
})

hist(pvals_null, breaks = 40, col = "#2A9D8F",
     main = "p-values under H0 are uniform", xlab = "p-value")

# Simulate under H1
pvals_alt <- replicate(n_sims, {
  x <- rnorm(30, mean = 0.5, sd = 1)
  t.test(x, mu = 0)$p.value
})

hist(pvals_alt, breaks = 40, col = "#F4A261",
     main = "p-values under H1 pile near zero", xlab = "p-value")

# Empirical power at alpha = 0.05
mean(pvals_alt < 0.05)
power.t.test(n = 30, delta = 0.5, sd = 1)$power
```

Output & Results

```
[1] 0.763    # empirical power
[1] 0.752    # theoretical power
```

Under H_0, the histogram is flat. Under H_1, p-values pile near zero. The empirical rejection rate at $\alpha = 0.05$ matches the theoretical power.

Interpretation

When reporting, pair p-values with effect sizes and confidence intervals. "The difference was 3.2 mmHg (95 % CI 0.6 to 5.8 mmHg, $p = 0.015$)" conveys both the magnitude and the statistical significance.

Practical Tips

- Never report a p-value alone; always include the effect size and its CI.
- Pre-specify one-sided vs. two-sided tests; switching after seeing data is fishing.
- Large samples make tiny effects statistically significant; focus on practical significance when n is large.
- Multiple testing inflates false-positive rates; adjust with Bonferroni, Holm, or FDR.
- "Nearly significant" ($0.05 < p < 0.10$) is not a meaningful category; report the exact value.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests